Homework 4
Minimum Enclosing Circle

Julia Reynolds, Akiva Lippert

October 17, 2026

Prompt

Submit a short report containing:

1. Your algorithm explanation
2. Time Complexity analysis
3. Improvement discussion
4. Computed circle
5. Three timing results

1 Algorithm

We selected Welzl's algorithm to implement. It randomly selects a point p from the set of all (current points) P, and calculates the minimum enclosing circle for the set of points $P - p$. If p is also on the boundary for the calculated circle, then the circle found is the minimum enclosing circle for all of P. If p is not on the boundary, then it recurses with R as a parameter, where R is the set of all points known to be on the boundary of the current circle. [1]

2 Time Complexity

One of the reasons we selected Welzl's algorithm is because it's a linear-time algorithm. It's tail-recursive but not branching, so each recursion adds a constant amount of work, and it will only recurse up to the number of points in P. This leads to $O(n)$ time complexity. [1]

3 Improvement

There is a small optimization to be made with Welzl's algorithm that we did not implement. The optimization is to store the set of points P as a global variable, then queue any points that fall outside of a given recursion's $P - p$ calculations to be recursed into earlier in the algorithm, removing some randomness from the selection of the next p in favor of early termination. [1]

4 Computed vs. AI

Compared to the AI's response in the homework prompt, we reached the same conclusion that the minimum enclosing circle for the 500 points in `random_2d_points_500.txt` has a center at $962.5, 1018.5$ and a radius of about 1365.58 units.

5 Program Runs

```
Attempting to read files from ../Graphs

2d_points_3.txt               -      3 coordinates
random_2d_points_500.txt      -    500 coordinates
square.txt                    -      4 coordinates

Read 3 files

Attempting to find minimum enclosing circle for each file.

Results:

---
File: 2d_points_3.txt

Runtimes:
        File Read Time   - 7275500 ns
        Find Circle Time - 1507600 ns

Circle:
        Center: (4.5, 4.5)
        Radius: 3.535533905029297

---
File: random_2d_points_500.txt

Runtimes:
        File Read Time   - 220100 ns
        Find Circle Time - 2645700 ns

Circle:
        Center: (962.5, 1018.5)
        Radius: 1365.5880126953125

---
File: square.txt

Runtimes:
        File Read Time   - 52100 ns
        Find Circle Time - 11100 ns

Circle:
        Center: (1.5, 1.5)
        Radius: 0.7071067690849304
```

References

[1] Wikipedia contributors. Smallest circle problem: Welzl's algorithm. *Wikipedia*, 2026.